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RESEARCH ARTICLE

PFOS and PFOA in paired urine and blood from general adults and pregnant women: assessment of urinary elimination

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Abstract Although levels of perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA) in human blood are well documented, information on elimination of these chemicals is limited. In this study, PFOS and PFOA were analyzed in 81 whole blood–urine paired samples from general adults and pregnant women in Tianjin, China. PFOS and PFOA were detected in 48 and 76 % of adult urine (AU) samples, with geometric mean (GM) concentrations of 0.011 and 0.008 ng/mL, respectively; whereas relatively low PFOS and PFOA concentrations were found in maternal urine (MU) samples, with GM concentrations of 0.006 and 0.003 ng/mL, respectively. For PFOA, the coefficients of Pearson's correlation between whole blood concentrations and creatinine-adjusted and creatinine-unadjusted urinary concentrations were 0.348 (p=0.013) and 0.417 (p=0.002), respectively. The GM urinary elimination rates of PFOS ($PFOS_{UER}$) and PFOA ($PFOA_{UER}$) were 16 and 25 %, respectively, for adults. These results indicate that urine is an important pathway of excretion of perfluoroalkyl substances (PFASs). The partitioning ratios of PFAS concentration between urine and whole blood ($PFAS_{U/B}$) in pregnant women ($PFOS_{U/B}$, 0.0004; $PFOA_{U/B}$, 0.0011) were significantly lower (p= 0.025 for $PFOS_{U/B}$, p=0.017 for $PFOA_{U/B}$) than the ratios found in non-pregnant women ($PFOS_{U/B}$, 0.0013; $PFOA_{U/B}$, 0.0028). Furthermore, our results suggest a clear gender difference in the urinary elimination of PFOA, with male adults (31 %) having significantly higher $PFOA_{UER}$ than that of female adults (19 %). $PFOS_{UER}$ was significantly inversely correlated with age (r=−0.334, p=0.015); these findings suggest that urinary elimination of PFOS is faster in young adults than in the elderly.

Keywords Perfluorooctane sulfonate · Perfluorooctanoic acid · Pregnant women · Urinary elimination rates

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Introduction

Perfluoroalkyl substances (PFASs) are a group of chemicals that have been widely used in many industrial and household products since the 1940s (Prevedouros et al. 2005). Widespread global distribution of PFASs, in human blood and breast milk, has been summarized (Houde et al. 2006, 2011; Lau et al. 2007; Post et al. 2012). Recent epidemiological investigations have shown that the two most commonly studied PFASs, perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA), were associated with potential adverse outcomes in fetal growth (Apelberg et al. 2007; Fei et al. 2007; Kim et al. 2011). Animal studies have indicated that PFOS and PFOA may disturb the fatty acid metabolism, affect the reproductive system, and induce adverse effects in liver and other tissues (Kennedy et al. 2004; Post et al. 2012; Seacat et al. 2002).

PFASs are not expected to accumulate in lipids but rather bind to proteins (Han et al. 2003; Jones et al. 2003). Furthermore, concerns over persistence and bioaccumulation were increased following the evidences of the long half-lives of PFOS (5.3 years) and PFOA (2.3 years) in humans (Bartell et al. 2010; Olsen et al. 2007). During recent years, many studies (Apelberg et al. 2007; Beesoon et al. 2011; Fromme et al. 2007; Kim et al. 2011; So et al. 2006; Tao et al. 2008; Zhang et al. 2010) were conducted to assess human exposure to PFOS and PFOA by detecting both of chemicals in tissues of fetus, infants, children, and adults. Nevertheless, in addition to exposure, PFOS and PFOA elimination from human body is another important aspect to influence their contamination level in human, and this is necessary data for assessing the human health risk from PFASs exposure. Urine is an important excretion pathway for human metabolism wastes and toxicant pollutants due to the fact that low urinary levels (<1.0 ng/mL) of PFOS and PFOA have been reported in adults (Harada et al. 2005; Perez et al. 2012; Zhang et al. 2013b; Zhou et al. 2014).

However, very few studies explored the occurrence of PFASs in paired samples of blood and urine (Harada et al. 2005; Zhang et al. 2013b; Zhou et al. 2014). Harada et al. (2005) measured PFOA and PFOS concentrations in 20 paired serum and urine samples collected from adults in Japan, and renal clearance rates (CL_{renal}) ranged from 0.11 to 1.15 mL day^{-1} kg^{-1} for PFOS and 0.75 to 3.33 mL day^{-1} kg^{-1} for PFOA. The CL_{renal} of PFOS and PFOA calculated for Chinese adults (Zhang et al. 2013b; Zhou et al. 2014) were in the same range as those reported in Japan (Harada et al. 2005). However, the contribution of urine to elimination of PFASs is still not well understood.

Furthermore, little is known on the differences in PFOS and PFOA elimination between non-pregnant women and pregnant women. Because the developing fetus is more sensitive to the effects of environmental chemicals than are adults (Stillerman et al. 2008; Woodruff et al. 2011), an understanding of excretion of PFOS and PFOA in pregnant women would provide information on the prenatal exposure to these chemicals.

In this study, we determined PFOS and PFOA in paired whole blood and urine samples collected from adults in the general population as well as pregnant women. Associations of PFOS and PFOA concentrations between urine and whole blood and aged (21–40 vs. 41–50 vs. 51–62 years), gender (males vs. females), and cohort (pregnant women vs. non-pregnant women) differences in urinary elimination rates were examined.

Materials and methods

Whole blood samples were extracted by ion-pair extraction method as described earlier (Hansen et al. 2001); urine samples were extracted using Oasis WAX SPE cartridge (Waters Corp, Milford, MA, USA) and were purified using Envi-carbon cartridge (Supelco, Inc., Bellefonte, PA, USA). Concentrations of PFOS and PFOA were determined with Waters Acquity UPLC equipped with Waters Acquity TQD triple quadrupole mass spectrometer (UPLC-MS/MS). Ten microliters of urine was diluted with milli-Q water (160-fold), and 800 ng of creatinine-d_3 was added as an internal standard. Creatinine levels in adult urine were analyzed by UPLC-MS/MS, as reported earlier (Liao and Kannan 2012).

Whole blood (n=5) and urine (n=5) samples were selected randomly, and 10 ng of PFOS and PFOA (10 μL, 1.0 ng/μL) as well as internal standards (MPFOS and M8PFOA; 10 μL, 1.0 ng/μL) were spiked and passed through the entire analytical procedure. Mean matrix-spike recoveries of PFOS and PFOA were 95±8 and 119±5 % in whole blood and 81±10 and 66±5 % in urine, respectively. Mean recoveries of MPFOS and M8PFOA were 95±8 and 100±9 % in whole blood and 68±6 and 73±12 % in urine, respectively. Details regarding reagents and chemicals, sample preparation, instrumental analysis, quality assurance, and quality control are given in the Electronic supplementary material.

Study participants and samples preparation

The sample collection and analysis were approved by the Institutional Review Board of Nankai University, China. A total of 81 paired whole blood and urine samples from the general adult population (n=54 pairs) and pregnant women (n=27 pairs) were collected in Tianjin, China, from May 2010 to June 2010. All pregnant women were randomly recruited from a hospital, and the adults were randomly recruited from local families. All participants were healthy, and none reported occupational exposure to PFASs. The average (range) ages of adults and pregnant women were 43.2 years (22–62) and 30.6 years (21–39), respectively. Overall, 29 of 54 adults were males; 82 % of pregnant women were primiparities, and 18 % had given birth to their second child. Detailed demographic information for the donors is provided in Table S1.

All adult blood (AB) and adult urine (AU) samples were first morning voids (fasting time, >8 h). The paired maternal whole blood (MB) and maternal urine (MU) samples were collected in a preoperative holding area within an hour after delivery. Urine samples were collected in 50 mL polypropylene tubes. Whole blood samples were obtained in heparin vacutainer connected directly to glass barrel syringes. All of the collected whole blood and urine specimens were frozen at −20 °C until analysis.

Calculation of urinary elimination

Based on measured PFAS (i.e., PFOS and PFOA) concentrations in whole blood-urine pairs, daily urinary elimination

($PFAS_{UE}$, ng/day), total daily intake ($PFAS_{TDI}$, ng/day), and urinary elimination rates ($PFAS_{UER}$, %) of PFOS and PFOA were estimated by using models described as below:

(1). For $PFAS_{UE}$,

$$PFAS_{UE} = V_U \times C_U$$

where C_U is the PFAS concentration in urine (ng/mL) and V_U is the daily urine excretion volume (mL/day). Individual values of urinary PFAS concentration from each donor were used for the calculation; an outlier value of 0.296 ng PFOS/mL in AU was excluded from the calculation of $PFOS_{UE}$. Values of V_U of individual normal adults may vary, and range from 1000 to 2000 mL, which is affected by sex, body weight, and weather conditions; however, these exact values of each individual could not be obtained. Hence, the average V_U (non-pregnant women and pregnant women, 1200 mL/day; males, 1600 mL/day) reported by the International Commission on Radiological Protection (2003) were used in this study.

(2). For the calculation of $PFAS_{TDI}$, a first-order pharmacokinetic (PK) model that has been widely used in previous studies (Lorber and Egeghy 2011; Thompson et al. 2010; Zhang et al. 2010) was employed.

$$PFAS_{TDI} = \frac{0.693}{t_{1/2}} \times V_D \times C'_B \times W$$

where $t_{1/2}$ is the half-life of PFAS in human serum (day), V_D is the apparent volume of distribution (mL/kg), C'_B is the serum PFAS concentrations in adults (ng/mL), and W is the body weight (kg). The parameters for this model were estimated based on serum data (Lorber and Egeghy 2011; Thompson et al. 2010; Zhang et al. 2010), therefore, C'_B was used for calculation of $PFAS_{TDI}$. The whole blood data were converted to a serum (i.e., C'_B) basis by multiplying by a factor of 2 (Spliethoff et al. 2008). For PFOS and PFOA, average $t_{1/2}$ values were estimated to be 1791 days (i.e., 5.4 years) and 840 days (i.e., 2.3 years), respectively, which were taken from recent studies examining the declining human serum concentrations (Bartell et al. 2010; Olsen et al. 2007). In accordance with Thompson et al. (2010) and Lorber and Egeghy (2011), we used V_D (PFOS, 230 mL/kg; PFOA, 170 mL/kg) values calibrated from human serum and exposure data. For non-pregnant women aged 21–50 years, the total clearance was corrected by menstrual blood loss (42 mL/month) (Harada et al. 2005). For body weights, 55 and 63 kg were used for female (i.e., non-pregnant women) and male adults, respectively (Yang et al. 2005).

Variability and uncertainties in the PK model are associated with model parameters. The V_D used in previous PFOS and PFOA modeling studies was obtained from animal dosing studies, which range from as low as 140 to as high as 6000 mL/day (Lorber and Egeghy 2011). Species-specific differences in kinetics of PFOS and PFOA have been reported (Tatum-Gibbs et al. 2011); therefore, parameters such as V_D and $t_{1/2}$ determined from human studies are necessary for human exposure assessments. Furthermore, the PK model is considered only valid for steady-state conditions (Fromme et al. 2007; Thompson et al. 2010). No age-related accumulation of PFOS (r=0.257, p=0.063) and PFOA (r=0.198, p=0.156) was found for general adults, and therefore we assumed a steady-state condition in adults. $PFAS_{TDI}$ could not be estimated for pregnant women because blood PFASs levels vary during pregnancy (Javins et al. 2013).

(3). For $PFAS_{UER}$,

$$PFAS_{UER} = \frac{100 \times PFAS_{UE}}{PFAS_{TDI}}$$

where $PFAS_{UE}$ is daily urinary elimination (ng/day) and $PFAS_{TDI}$ is the total daily intake modeled by the PK model (ng/day).

Statistical analysis

Due to the variability in the volume of spot urine samples (that can influence concentrations of contaminants), adjustment of urinary data to creatinine content is performed to correct for variable volumes. That is, the concentration of a contaminant in urine (volume based concentration, μg/mL) is divided by the creatinine concentration (g/mL), to provide creatinine-adjusted concentration (μg/g).

Data analysis was performed with SPSS, Version 17.0. Concentrations, $PFAS_{UE}$, $PFAS_{TDI}$, and $PFAS_{UER}$ were tested for normality using the Kolmogorov–Smirnov test. Differences between groups were compared by one-way ANOVA when the two sets of data were normally distributed; otherwise, Mann–Whitney U test was used. Pearson's correlation coefficients were used for the analysis of relationship between two sets of data with normal distribution (or log-normal distribution); otherwise, the Spearman's rank correlation coefficient was used. A value of $p < 0.05$ denoted significance.

Results and discussion

Concentrations (range, median, geometric mean (GM), and mean) of PFOS and PFOA in whole blood-urine pairs collected from adults and pregnant women in Tianjin, China, are shown in Table 1. Non-detected observations (<LOQ) are assigned a value equal to half of their corresponding LOQs.

Urinary PFOS and PFOA concentrations

The number of AU samples with detectable concentrations of PFOS and PFOA differed (PFOA n=41; PFOS n=26) (Table 1). The GM PFOS and PFOA concentrations in AU were 0.011 and 0.008 ng/mL, respectively. PFOA and PFOS were detected in 30 and 11 % of the 27 MU samples, with GM concentrations of 0.003 and 0.006 ng/mL, respectively (Table 1). The mean urinary concentrations of PFOS (0.023 ng/mL) in adults from this study (location, Tianjin City) were lower than the concentrations (mean, 0.050 ng/mL) reported earlier from Shanxi Province in China (Li et al. 2013), while urinary PFOA concentration in Tianjin (mean, 0.011 ng/mL) were similar to those reported in Shanxi (mean, 0.013 ng/mL) (Li et al. 2013). On the contrary, whole blood concentrations of PFOS (mean, 12.8 ng/mL) and PFOA (mean, 2.71 ng/mL), found in the present study, were higher than the concentrations reported from Shanxi (mean serum levels, 8.65 ng PFOS/mL, 1.29 ng PFOA/mL) (Li et al. 2013). In comparison with the contemporaneous studies from other countries (Perez et al. 2012), urinary PFOS concentrations found in adults in this study are similar to the concentrations reported in Spain (<0.37 ng/mL in all samples); however, urinary levels of PFOA in Chinese adults were less than the concentrations reported in Spain (mean, 0.824 ng/mL).

Table 1 Concentrations (ng/mL for whole blood, ng/mL, or μg/g creatinine for urine) of PFOS and PFOA in whole blood–urine pairs from an adult population and pregnant women in China

	DR (%)	GM	Median	Mean	Min	Max
Adult blood (n=54)						
PFOS	100	8.62	10.3	12.8	0.7	65.5
PFOA	100	2.47	2.47	2.71	1.02	6.66
Adult urine (n=54, on a volume basis)						
PFOS	48	0.011	<LOQ	0.023	<LOQ	0.296
PFOA	76	0.008	0.009	0.011	<LOQ	0.041
Adult urine[a] (n=54, adjusted by creatinine)						
PFOS	48	0.007	<LOQ	0.031	<LOQ	1.125
PFOA	76	0.004	0.003	0.005	<LOQ	0.020
Maternal blood[b] (n=27)						
PFOS	100	13.4	12.4	14.7	7.39	36.1
PFOA	100	3.09	2.96	3.36	1.17	8.94
Maternal urine[c] (n=27, on a volume basis)						
PFOS	11	0.006	<LOQ	0.015	<LOQ	0.132
PFOA	30	0.003	<LOQ	0.006	<LOQ	0.030

DR detection rate, *GM* geometric mean value, *min* minimum value, *max* maximum value, *<LOQ* concentration value lower than LOQ

[a] Creatinine-adjusted concentration (μg/g Cr)

[b] PFAS concentrations in MB were discussed in another study for comparison of PFASs among MB, cord blood, amniotic fluid, and placenta (Zhang et al. 2013a)

[c] The concentrations of creatinine was not analyzed in maternal urine, therefore, creatinine-adjusted urinary PFAS concentrations were not calculated

Correlation between paired urine and blood

The Pearson's correlation test showed that creatinine-unadjusted (r=0.417, p=0.002; Fig. 1a) and creatinine-adjusted (r=0.348, p=0.013; Fig. 1b) PFOA concentrations in AU were significantly correlated with PFOA concentrations in AB. AU PFOS concentrations on the volume basis (i.e., creatinine-unadjusted) were marginally correlated (r=0.254, p=0.067) with AB PFOS concentrations (Fig. S1a), while the relationship between creatinine-adjusted PFOS concentrations in AU and PFOS concentrations in AB was not significant (r=0.237, p=0.105) (Fig. S1b). It is worth noting

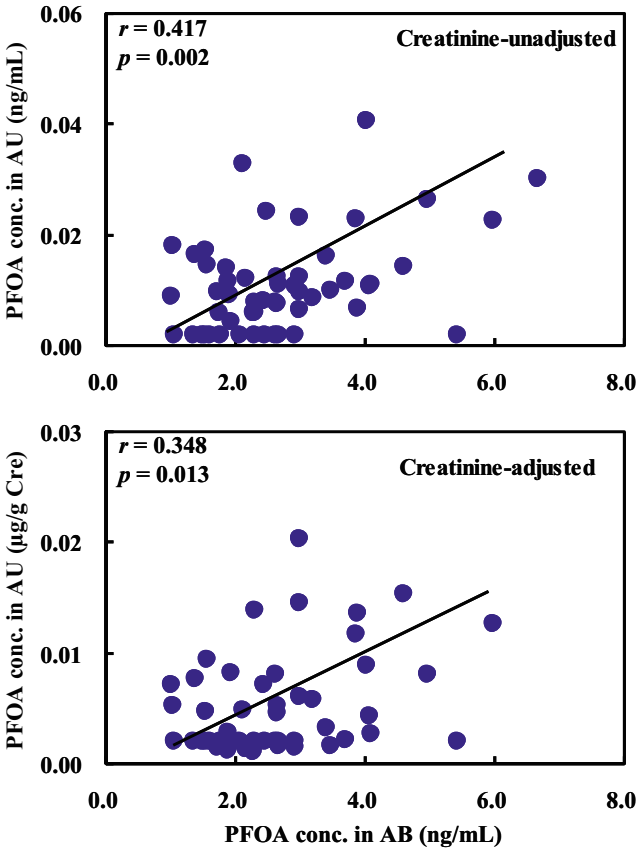

Fig. 1 Correlations of PFOA concentrations between paired whole blood and urine from adults in China. **a** PFOA concentrations in urine were volume based (ng/mL); **b** PFOA concentrations were creatinine adjusted (μg/g creatinine)

that relatively low detection rates of PFOS (48 %) found in AU could affect the results of association of PFOS concentration between urine and whole blood. Our results are inconsistent with a previous study reporting a significant correlation of PFOS concentrations between urine and serum from Chinese adults, whereas the significant correlation was not found for PFOA (Li et al. 2013). Although there is controversy about the creatinine adjustment of urinary concentrations, our results suggest that urine is a potential matrix for monitoring human body burdens of PFOA but that further investigations with greater number of donors are needed.

PFAS concentrations between urine and whole blood were not significantly correlated in pregnant women, which could be due to the low detection rates of PFASs obtained in the MU (30 % for PFOA and 11 % for PFOS). However, a statistically significant relationship (r=0.724, p=0.042) was found in those samples (n=8 pairs) that had measurable concentrations of PFOA (>LOQ) in both urine and blood.

Urinary elimination and daily intake of PFOS and PFOA

Among all general adults, the GM $PFOS_{UE}$ and $PFOA_{UE}$ were estimated to be 14.0 and 11.0 ng/day, respectively (without creatinine adjustment) (Figs. 2c and 3c). The $PFAS_{UE}$ of PFOS and PFOA were calculated to be 6.91 and 4.16 ng/day, respectively, for all pregnant women (Figs. 2c and 3c). Our data on $PFOS_{UE}$ and $PFOA_{UE}$ are comparable with those found for adults from Japan (PFOS, 1.4–36.2 ng/day; PFOA, 8.7–39.0 ng/day) (Harada et al. 2005).

The GM values for $PFOS_{TDI}$ and $PFOA_{TDI}$ were estimated to be 89.2 and 44.2 ng/day, respectively (Figs. 2d and 3d), for adults. The maximum $PFOS_{TDI}$ (513 ng/day or 0.012 μg kg^{-1} day^{-1}) and $PFOA_{TDI}$ (118 ng day^{-1} or 0.0018 μg kg^{-1} day^{-1}) values for the adults were two times and 2 orders of magnitude, respectively, lower than the oral reference dose for non-carcinogenic effects (PFOS, 0.025 μg kg^{-1} day^{-1}, PFOA: 0.333 μg kg^{-1} day^{-1}) (Thayer 2002).

The GM values for $PFOS_{UER}$ and $PFOA_{UER}$ were 16 and 25 % for adults (Figs. 2e and 3e). These results indicate that urine is an important excretion pathway for PFASs in humans. Other elimination pathways of PFASs such as feces and biliary excretion has been reported in several animal studies (Kudo et al. 2001; Ohmori et al. 2003); feces was considered to be a minor elimination pathway for PFOS and PFOA (Cui et al. 2010; Kudo et al. 2001).

Pregnant women vs. non-pregnant women

In this study, $PFAS_{TDI}$ could not be estimated for pregnant women, therefore, $PFAS_{UER}$ were not calculated for pregnant women. To assess the differences in urinary elimination of PFASs between pregnant women and non-pregnant women, we calculated the partitioning ratios of PFAS concentration between urine and whole blood ($PFAS_{U/B}$) for pregnant women and non-pregnant women, respectively. Pregnant women (0.0004) show significantly (p=0.025) lower $PFOS_{U/B}$ than non-pregnant women (0.0013) (Fig. 1). Furthermore, $PFOA_{U/B}$ value calculated for pregnant women (0.0011) was also significantly lower (p=0.017) than that found for non-pregnant women (0.0028). Furthermore, although no significant differences in $PFOS_{U/B}$ (p=0.166) or $PFOA_{U/B}$ (p=0.229) were found between full-term pregnant women (gestational age, ≥40 weeks) and non-full-term pregnant women (gestational age, <40 weeks), full-term pregnant women (0.0004 for PFOS, 0.0009 for PFOA) had lower $PFOS_{U/B}$ and $PFOA_{U/B}$ than non-full-term pregnant women (0.0006 for PFOS, 0.0015 for PFOA) did. These results indicate slower clearance rates of PFASs in pregnant women than in non-pregnant women. However, blood volume in pregnant women is increased with increasing gestational age, while urine volume is unchanged during pregnancy (ICRP 2002), therefore, it should be noted that unadjusted $PFAS_{U/B}$ may cause uncertainties in the comparison of $PFAS_{U/B}$ between pregnant women and non-pregnant women.

In animal studies (Kudo et al. 2001; Lee and Schultz 2010; Tatum-Gibbs et al. 2011), rats treated with testosterone had lower $PFOA_{UER}$; treatment of female minnows with synthetic androgen, trenbolone, substantially delayed the elimination of PFOA. These results suggest that sex hormones exert an inhibitory effect on urinary elimination of PFASs in rats and fathead minnows (Lee and Schultz 2010; Tatum-Gibbs et al. 2011). Besides, the lower $PFAS_{UER}$ may also be due to the fact that the PFAS in pregnant women has other elimination pathway, such as to placenta and cord blood. In this study, therefore, all non-pregnant women were divided into two groups: ages 21–50 (i.e., menstruation group) and 51–62 years (i.e., menopause group) according to menstrual status. The order of values calculated for $PFAS_{U/B}$ were pregnant women (0.0004) <non-pregnant women aged 51–62 years (0.0006)<all non-pregnant women (0.0013)<non-pregnant women aged 20–50 years (0.0018) for PFOS, and pregnant women (0.0011) <non-pregnant women aged 51–62 years (0.0028)≈all non-pregnant women (0.0028)≈non-pregnant women aged 20–50 years (0.0029) for PFOA. Hence, we deduced that the lower $PFAS_{U/B}$ found in pregnant women than those in non-pregnant women suggest differences in sex hormone levels (e.g., estrogen, chorionic gonadotropin, hCG, and progesterone hormones), which are higher during pregnancy.

Gender- and age-related differences in urinary elimination rates

Gender-related differences in $PFAS_{UER}$ were examined. The values for $PFOA_{UER}$ in male adults (31 %) were significantly higher (p=0.002) than those in female adults (i.e., non-

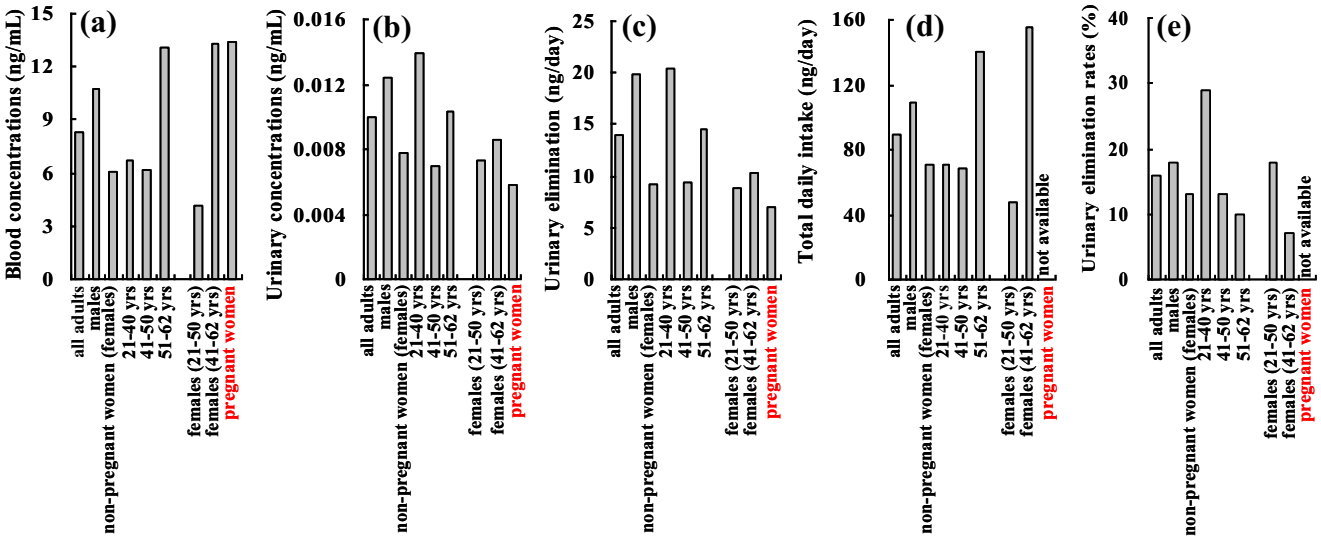

Fig. 2 Geometric mean values of whole blood concentrations (**a**), urinary concentrations (**b**), daily urinary elimination (**c**), modeled total daily intake (**d**), and urinary elimination rates (**e**) of PFOS for all participants, stratified according to gender and age. Total daily intake of PFOS was not estimated for pregnant women due to varying blood PFOS levels during pregnancy (Javins et al. 2013). Outlier concentration values (0.289 ng/mL in urine, 65.5 ng/mL in blood) were excluded from data analysis

pregnant women, 19 %) (Fig. 3e). The results indicated that the urinary elimination of PFOA in adult females was slower than that found for adult males. Males had slightly higher $PFOS_{UER}$ than those of females. However, no significant difference was found in $PFOS_{UER}$ between adult male and female donors (Fig. 2e).

Influence of age of adults on $PFAS_{UER}$ were examined by Pearson's correlation coefficients. No significant associations between age and $PFOA_{UER}$ were found across all general adults. However, $PFOS_{UER}$ was significantly inversely correlated with age ($r=-0.334$ and $p=0.015$). Although the number of samples is small, the observed relationship between age and $PFOS_{UER}$ suggests that urinary elimination of PFOS was faster in young adults than that in the elderly. Furthermore, we categorized the whole blood-urine pairs from all general adults into three groups: 21–40, 41–50, and 51–62 years. Among all adults, the order of $PFOS_{UER}$ in each age groups was 21–40 years (29 %)>41–50 years (13 %)>51–62 years (10 %) (Fig. 2e). Although the relationships between age and $PFOA_{UER}$ were not found, the highest $PFOA_{UER}$ was found for the age group of 21–40 years (32 %), followed by 41–50 years (24 %) and 51–62 years (19 %) (Fig. 3e).

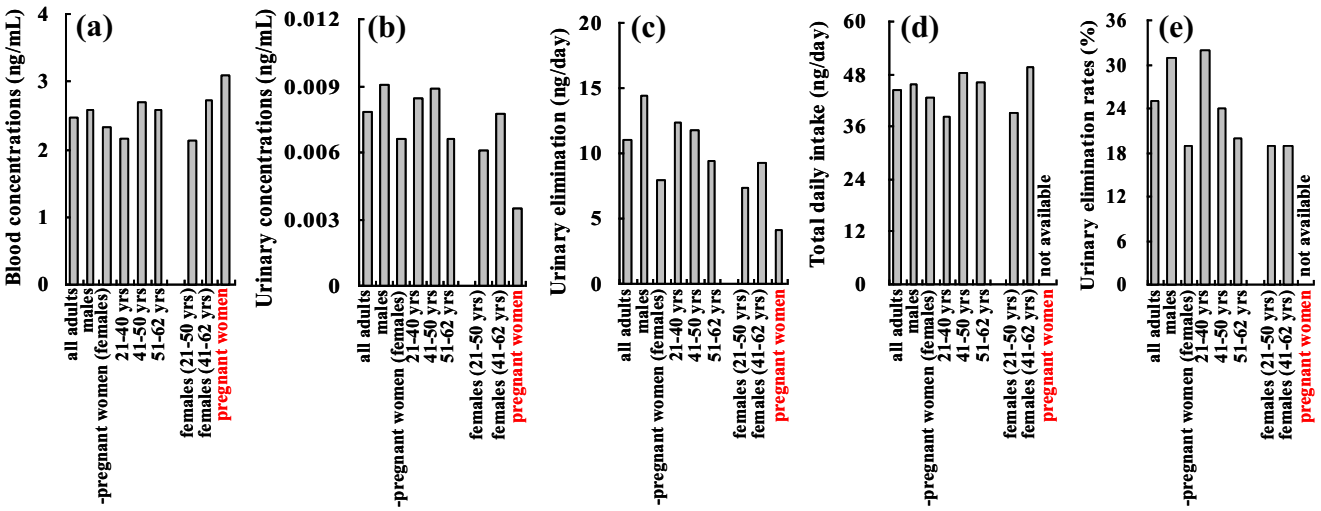

Fig. 3 Geometric mean values of whole blood concentrations (**a**), urinary concentrations (**b**), daily urinary elimination (**c**), modeled total daily intake (**d**), and urinary elimination rates (**e**) of PFOA for all participants, stratified according to gender and age. Total daily intake of PFOA was not estimated for pregnant women due to varying levels of blood PFOA during pregnancy (Javins et al. 2013)

Conclusions

This study showed urinary PFOS and PFOA concentrations were 3 orders of magnitude lower than whole blood concentrations in adults and pregnant women. The GM $PFOS_{UER}$ and $PFOA_{UER}$ were 16 and 25 %, respectively, for general adults, which indicated urine is an important excretion pathway for PFASs in humans. Pregnant women had significantly lower $PFOA_{U/B}$ and $PFOS_{U/B}$ than non-pregnant women. Adult females had significantly (p=0.002) lower $PFOA_{UER}$ than adult males, and $PFOS_{UER}$ decreased with age (r=−0.334 and p=0.015).

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Conflict of interest There is no conflict of interest.

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